ATTERY INPUT + PROTECTION

BATTERY_XT60_PIGTAIL

External latching E-stop / disconnect goes between F1 and VBAT_RAW.

5V CONTROL BUCK (ESP32 + LOGIC ONLY)

1000uF 25V LOW_ESR

100nF ADC FILTER

MP1584EN-LF-Z

WHEEL ENCODERS (3.3V)

220uF 25V LOW_ESR

330uF 10V LOW_ESR

OWER PASS-THROUGH (SEPARATE 5V/5A REGULATOR)

PI_5V_5A_INPUT

PI_POWER_OUT_2x5V_2xGND

1000uF 10V LOW_ESR

t Jr / screw terminal, not ordinary 0.1in headers, for Pi 5 A power.

SP32-PICO-KIT V4.1 SOCKETS

ESP32_PICO_KIT_V4.1_CARRIER

provisional carrier pad pattern; verify against the real board before ordering.

RASPBERRY PI UART / CONTROL

PI_UART_CONTROL

UART is 3.3V: Pi GPIO14 TX -> ESP RX0; ESP TX0 -> Pi GPIO15 RX.

Do not connect Pi/ESP UART while Micro-USB serial is actively driving RX0/TX0.

LEFT_ENCODER

RIGHT_ENCODER

74LVC1G14

1nF FILTER

Encoder inputs include pull-ups, RC filtering, and Schmitt buffers.

SAFETY / REVIEW NOTES

1. VNH5019 exposed pads need reflow, thermal vias, and large copper pours.
2. Verify actual motor stall current before ordering production PCB.
3. Pi power is NOT made by U1. Feed J8 from a separate regulated 5V/5A supply.
4. Never power ESP32 from USB and EXT_5V simultaneously.
5. External fuse and latching E-stop are mandatory.

LEFT MOTOR DRIVER

VNH5019A-E

LEFT_MOTOR

470uF 25V LOW_ESR

10nF CURRENT FILTER

RIGHT MOTOR DRIVER

VNH5019A-E

RIGHT_MOTOR

470uF 25V LOW_ESR

10nF CURRENT FILTER

3-DOF ARM SERVO CONTROL

15A_EXTERNAL_SERVO_FUSE

SMBJ8.0A

2200uF 10V LOW_ESR

SERVO_6V_10A_INPUT

PCA9685PW

MAIN_SWING_25KG_SERVO

ELBOW_MG996R_SERVO

WRIST_MG90S_SERVO

Servo pin order: 1 GND, 2 regulated 6V, 3 PWM. Verify cable colors before plugging in.

Pi power is a separate 5 V / 5 A pass-through branch
DRAFT: verify motor stall current and footprints before manufacture
3S LiPo; 2x GA25-370; provisional 5 A stall per motor

Harrison

Sheet: /ELBOW_MG996R_SERVO/
File: pickleball_robot_controller.kicad_sch

Title: Pickleball Robot Wheel Controller + Raspberry Pi Interface

Size: A3	Date: 2026-07-12	**Rev: R1-DRAFT**
KiCad E.D.A. 10.0.4		Id: 1/1